SAGARKAVACH-AI: AUTONOMOUS MARITIME SURVEILLANCE AND NAVAL THREAT INTELLIGENCE SYSTEM

PROJECT REPORT

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ABSTRACT

The growing importance of maritime surveillance and ocean monitoring has created a demand for automated systems capable of detecting and classifying naval vessels in images and video streams. Traditional methods based on manual observation or rule-based approaches often fail to achieve reliable performance due to variations in ship size, orientation, environmental conditions, and complex backgrounds.

This project presents a computer vision-based approach for naval ship detection and classification using deep learning techniques. A custom dataset was constructed by combining images from multiple sources, including partially labeled datasets and unlabeled external collections. To address inconsistencies in annotation, the dataset was unified and labeled using Roboflow with the assistance of the Segment Anything Model (SAM), ensuring accurate and consistent bounding box annotations across all classes.

The system employs a YOLOv8-based object detection model with oriented bounding box (OBB) capability to effectively detect ships with varying orientations. The model is trained on the processed dataset and integrated into a video inference pipeline for real-time detection.

The proposed approach replaces traditional heuristic-based methods with a data-driven learning framework, improving detection reliability and enabling practical deployment in real-world maritime scenarios. This work contributes to the development of efficient and scalable solutions for automated naval surveillance and monitoring applications.

KEYWORDS: Naval Ship Detection, YOLOv8, Oriented Bounding Boxes, Roboflow, Segment Anything Model, Deep Learning, Object Detection, Computer Vision, Maritime Surveillance, Video-Based Detection

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
OBB	Oriented Bounding Box
YOLO	You Only Look Once
EDA	Exploratory Data Analysis
SAM	Segment Anything Model
mAP	Mean Average Precision
IoU	Intersection over Union
FPN	Feature Pyramid Network
CNN	Convolutional Neural Network
FPS	Frames Per Second
CV	Computer Vision
GPU	Graphics Processing Unit
HSV	Hue Saturation Value
DDG	Destroyer (Guided Missile Destroyer)
CVN	Aircraft Carrier (Nuclear Powered)
CVT	Corvette Class Vessel
SSN	Submarine (Nuclear Powered)

1. INTRODUCTION

1.1 Background

Maritime surveillance plays a critical role in modern defense and security operations, particularly in monitoring naval activities across territorial and international waters. With the rapid increase in satellite imagery, aerial reconnaissance, and unmanned surveillance platforms, the volume of visual maritime data has grown significantly. Manual monitoring of such large-scale data is time-consuming, labor-intensive, and prone to human error. Consequently, there is an increasing need for automated, intelligent systems capable of detecting and classifying naval vessels efficiently and accurately.

Recent advancements in deep learning and computer vision have enabled the development of automated object detection systems capable of identifying complex objects in challenging environments. Among these, the YOLO (You Only Look Once) family of models has emerged as one of the most effective real-time object detection architectures due to its balance of speed and accuracy. In particular, YOLOv8 introduces improved detection performance along with support for oriented bounding box (OBB) detection, making it well-suited for aerial and satellite imagery where objects often appear at arbitrary orientations.

Naval vessels present unique detection challenges, including varying scales, complex backgrounds, occlusions, and rotational variations in aerial imagery. Traditional axis-aligned bounding boxes often fail to accurately capture elongated objects such as destroyers and submarines when viewed at angles. To address this limitation, this project utilizes YOLOv8-OBB, which incorporates oriented bounding boxes to improve localization accuracy for rotated naval vessels.

This project proposes **SagarKavach-AI**, an intelligent maritime surveillance system designed for automated detection and classification of naval warships. The system is trained on a custom dataset comprising four classes: aircraft carriers, corvettes, destroyers, and submarines. The trained model is further integrated into a real-time video inference pipeline for dynamic maritime monitoring.

The proposed system aims to enhance maritime situational awareness, reduce manual monitoring effort, and provide a scalable solution for defense, coastal surveillance, and naval intelligence applications.

1.2 Dataset Provenance and Preprocessing

The dataset used for SagarKavach-AI was self-curated from multiple heterogeneous sources to ensure diversity, robustness, and real-world applicability. A total of approximately four thousand naval warship images were collected across four classes: Aircraft Carrier, Corvette, Destroyer, and Submarine.

To ensure dataset diversity and reduce bias, images were sourced from the following channels:

- 50% — Web Scraping and Naval Warship Videos
  A significant portion of the dataset was obtained through web scraping from publicly available naval imagery and extracting frames from naval warship videos. Video frame extraction enabled capturing vessels from multiple angles, lighting conditions, and distances, improving model generalization.

- 35% — Open-Source Repositories
  Images were collected from publicly available datasets such as aerial imagery repositories, satellite imagery datasets, and open-source naval image collections.
- 15% — Licensed Datasets
  A smaller portion of the dataset was obtained from licensed datasets and curated collections to ensure inclusion of high-quality annotated naval imagery.

This multi-source data collection strategy ensured high variability in ship orientation, scale, weather conditions, and background clutter, making the dataset suitable for real-world maritime surveillance scenarios.

All collected images were annotated using Roboflow, which provided an efficient platform for dataset management, annotation, and export. The annotation process involved:

- Class labeling into four categories:
  - Aircraft Carrier
  - Corvette
  - Destroyer
  - Submarine
- Use of Oriented Bounding Box (OBB) annotations to handle rotational variance in aerial imagery
- Manual verification of annotations to ensure label correctness

Roboflow's annotation interface enabled accurate labeling and dataset version control, ensuring consistency across training, validation, and test sets. The annotated dataset was then exported in YOLOv8-OBB format for model training.

To improve model robustness and generalization, multiple preprocessing and augmentation techniques were applied. More than five preprocessing methods were used, including:

- Grayscale Conversion
  Images were converted to black-and-white to improve detection performance under low-color contrast conditions.
- Image Inversion
  Inverse transformations were applied to enhance detection capability under varying lighting conditions.
- Rotation and Orientation Augmentation
  Images were rotated at different angles to simulate aerial and satellite viewpoints.
- Scaling and Resizing
  Images were resized to 416 × 416 resolution for model training consistency.
- Horizontal and Vertical Flipping
  Flipping techniques were used to increase dataset diversity.
- Brightness and Contrast Adjustment
  Applied to simulate varying illumination conditions.
- Data Augmentation (Roboflow + YOLO Augmentation)
  Additional augmentations included:
  - Mosaic augmentation
  - HSV color jitter
  - Random scaling
  - Random cropping

These preprocessing techniques significantly improved the model's ability to detect naval vessels under diverse environmental conditions, thereby enhancing overall detection accuracy and robustness.

Figure 1: Sample images of Aircraft destroyer in our custom Naval Ship Dataset

1.3 Deep Learning and Object Detection Models

The detection and classification of naval warships in aerial and satellite imagery present significant technical challenges due to variations in scale, orientation, illumination, and background clutter. Traditional computer vision approaches based on handcrafted features such as Histogram of Oriented Gradients (HOG), Scale-Invariant Feature Transform (SIFT), and edge-based detection methods often fail to generalize effectively in complex maritime environments. These limitations have led to the widespread adoption of deep learning-based object detection architectures, which automatically learn hierarchical feature representations directly from raw image data. In this project, a deep learning-based object detection framework built upon the YOLOv8 (You Only Look Once Version 8) architecture is employed to achieve robust and real-time naval warship detection.

YOLOv8 represents a significant advancement in the YOLO family of single-stage detectors, offering improved feature extraction, enhanced training stability, and superior detection accuracy. Unlike two-stage detectors such as Faster R-CNN, which first generate region proposals and then perform classification, YOLOv8 formulates object detection as a unified regression problem. This enables the model to simultaneously predict bounding box coordinates, objectness scores, and class probabilities in a single forward pass, significantly reducing inference latency while maintaining high accuracy. This property is particularly advantageous for maritime surveillance applications where real-time processing of video feeds and satellite imagery is essential.

To further enhance detection capability in aerial and maritime imagery, this project employs the YOLOv8 Oriented Bounding Box (YOLOv8-OBB) variant. Naval vessels typically appear at arbitrary orientations in satellite and aerial imagery due to varying camera angles and ship movements. Conventional axis-aligned bounding boxes often fail to tightly encapsulate elongated objects such as destroyers and submarines, leading to inaccurate localization and reduced detection performance. The YOLOv8-OBB model addresses this limitation by incorporating rotated bounding box regression, enabling the prediction of object orientation in addition to position and scale. This results in improved localization accuracy and enhanced detection performance in complex maritime environments.

The selected model variant for this project is YOLOv8m-OBB (Medium Variant), which provides an optimal balance between computational efficiency and detection accuracy. The medium variant offers increased network depth and channel capacity compared to smaller variants, allowing the model to learn more discriminative features while remaining feasible for training on consumer-grade GPU hardware. The architecture consists of a CSPDarknet backbone for feature extraction, a Feature Pyramid Network (FPN) neck for multi-scale feature aggregation, and a decoupled detection head for

classification and localization tasks. This modular architecture enables the model to detect vessels of varying sizes and orientations effectively.

The model is trained on a custom-curated dataset containing multiple classes of naval vessels, including aircraft carriers, corvettes, destroyers, and submarines. During training, the model learns class-specific visual features such as hull geometry, deck structures, radar installations, and superstructure patterns. Data augmentation techniques such as rotation, scaling, and brightness variation further improve generalization capability, allowing the model to perform reliably under diverse environmental conditions including varying sea states, lighting conditions, and imaging perspectives.

The deep learning-based detection framework developed in this project incorporates the following key components:

- YOLOv8 Single-Stage Object Detection Architecture for Real-Time Performance
- YOLOv8 Oriented Bounding Box (OBB) for Rotationally Aware Detection
- YOLOv8m Model Variant for Optimal Accuracy–Performance Trade-off
- Multi-Scale Feature Extraction using CSPDarknet Backbone and FPN Neck
- Custom Naval Warship Dataset for Multi-Class Classification
- Advanced Data Augmentation for Robust Maritime Detection

The integration of these components enables the proposed system to achieve high detection accuracy while maintaining real-time inference capability. This deep learning-based approach provides a scalable and efficient solution for maritime surveillance, naval intelligence, and defense-oriented computer vision applications.

1.4 Dataset Preprocessing and Deployment Framework

Prior to model training, extensive dataset preprocessing is performed to ensure data consistency, improve model generalization, and enhance detection performance across diverse maritime environments. Raw images collected from multiple heterogeneous sources often contain variations in resolution, lighting conditions, orientation, and background complexity. Without appropriate preprocessing, these inconsistencies can negatively impact model convergence and reduce detection accuracy. Therefore, a structured preprocessing pipeline is implemented to standardize the dataset and prepare it for deep learning-based training using YOLOv8-OBB.

The preprocessing pipeline begins with image normalization and resizing, where all images are resized to a uniform resolution of 416 × 416 pixels to maintain consistency during training. This ensures that the model processes inputs of fixed dimensions, improving computational efficiency and stabilizing gradient updates. Pixel normalization is applied to scale image intensities into a standardized range, allowing the neural network to learn more effectively and preventing numerical instability during training.

In addition to basic preprocessing, extensive data augmentation techniques are applied to improve model robustness and generalization capability. Given that naval vessels may appear at varying orientations, scales, and lighting conditions, augmentation plays a crucial role in simulating real-world operational scenarios. Techniques such as rotation, flipping, brightness and contrast adjustment, grayscale conversion, image inversion, and HSV color transformations are applied to

increase dataset diversity. These transformations allow the model to learn invariant features and improve detection performance in challenging maritime environments such as low visibility, varying sea conditions, and satellite imagery distortions.

Furthermore, the dataset is organized into training, validation, and test subsets using a 70:20:10 split ratio. The training set is used for model learning, the validation set is used for hyperparameter tuning and performance monitoring, and the test set is reserved for final evaluation. This structured dataset partitioning helps prevent overfitting and ensures reliable performance assessment on unseen data. The dataset is also structured in YOLOv8-OBB format, with corresponding annotation files containing class labels and oriented bounding box coordinates for each image.

Following preprocessing and training, the trained model is integrated into a deployment and inference framework designed for real-world maritime surveillance applications. The deployment pipeline is implemented using OpenCV and the Ultralytics YOLOv8 inference engine, enabling efficient processing of video streams. The system sequentially processes video frames, performs object detection, and extracts predictions including class labels, confidence scores, and oriented bounding box coordinates.

Detected naval vessels are then visualized by overlaying rotated bounding boxes, class names, and confidence values directly onto each frame. This visualization provides intuitive and interpretable detection outputs for operators and analysts. The processed frames are subsequently compiled into an output video stream, enabling continuous monitoring and playback of detection results. The inference pipeline is optimized for real-time performance, allowing the system to process frames efficiently while maintaining high detection accuracy.

The dataset preprocessing and deployment framework incorporates the following key components:

- Image Resizing and Pixel Normalization for Training Consistency
- Multi-Method Data Augmentation for Robust Model Generalization
- Structured Dataset Splitting (Training, Validation, Test)
- YOLOv8-OBB Annotation Format for Rotational Object Detection
- Real-Time Video Inference Pipeline using OpenCV
- Bounding Box Visualization and Confidence-Based Detection Output

The integration of preprocessing and deployment components ensures that the proposed system operates efficiently in real-world maritime surveillance scenarios. This framework provides a scalable foundation for automated naval monitoring, coastal defense applications, and intelligent maritime threat detection systems.

1.5 Evaluation Metrics

To assess the performance of the proposed naval warship detection system, a comprehensive set of standard object detection evaluation metrics is employed. These metrics are widely adopted in computer vision research and provide quantitative insights into the model's classification accuracy, localization performance, and overall detection capability. Since object detection involves both identifying object categories and accurately localizing them within images, multiple complementary metrics are necessary to evaluate model effectiveness comprehensively.

The evaluation process is conducted on a held-out test dataset to ensure unbiased performance assessment. The trained YOLOv8-OBB model generates predictions consisting of class labels, confidence scores, and oriented bounding box coordinates. These predictions are then compared against ground truth annotations to compute performance metrics. The evaluation framework incorporates threshold-based confidence filtering and Intersection over Union (IoU) calculations to determine true positives, false positives, and false negatives, which form the basis for computing detection metrics.

Precision is used to measure the accuracy of the model's predictions. It represents the proportion of correctly detected objects among all predicted detections. High precision indicates that the model produces fewer false positives, meaning that when the model predicts a naval vessel, it is highly likely to be correct. Precision is particularly important in defense and surveillance applications, where false alarms may lead to unnecessary operational responses or resource allocation.

Recall evaluates the model's ability to detect all relevant objects within an image. It measures the proportion of actual objects that are correctly identified by the model. High recall ensures that fewer naval vessels are missed during detection, which is critical in maritime surveillance scenarios where undetected vessels could represent potential threats. Balancing recall with precision is essential to ensure both accurate and complete detection.

F1-Score provides a balanced evaluation metric by combining precision and recall into a single value. It is computed as the harmonic mean of precision and recall, making it particularly useful when both false positives and false negatives must be minimized. In multi-class naval warship detection, the F1-score helps assess overall detection reliability across all vessel categories.

Mean Average Precision (mAP) serves as the primary evaluation metric for object detection models. It measures detection performance across all classes by calculating the average precision at different recall levels. The mAP50 metric evaluates performance at an Intersection over Union (IoU) threshold of 0.50, while mAP50–95 computes the average precision across multiple IoU thresholds ranging from 0.50 to 0.95. These metrics provide a comprehensive understanding of both detection accuracy and localization precision.

Intersection over Union (IoU) is used to measure the overlap between predicted bounding boxes and ground truth annotations. A higher IoU value indicates better localization accuracy. For oriented bounding boxes, IoU calculation becomes particularly important, as the model must accurately capture both position and orientation of naval vessels in aerial imagery.

The evaluation framework incorporates the following key metrics:

- Precision for detection accuracy
- Recall for detection completeness
- F1-Score for balanced performance evaluation
- Mean Average Precision (mAP50 and mAP50–95) for overall detection performance
- Intersection over Union (IoU) for localization accuracy
- Confidence Threshold Analysis for optimal operating point selection

These evaluation metrics collectively provide a comprehensive assessment of the model's performance across classification accuracy, localization precision, and detection reliability. The use of multiple evaluation measures ensures that the proposed system is robust, reliable, and suitable for real-world maritime surveillance and naval threat detection application.

2. OBJECTIVES

The main objectives of this project are:

1. To develop an automated system for detecting and classifying naval ships in images and video streams using computer vision techniques.
2. To construct a unified and structured dataset by combining multiple sources and addressing inconsistencies in data organization and labeling.
3. To generate accurate and consistent annotations for all classes using Roboflow and the Segment Anything Model (SAM).
4. To perform exploratory data analysis (EDA) to understand class distribution, object characteristics, and dataset variability.
5. To implement an object detection model using the YOLOv8 framework with oriented bounding box (OBB) capability.
6. To train the model to accurately detect and localize different classes of naval ships, including aircraft carriers, corvettes, destroyers, and submarines.
7. To evaluate model performance using standard object detection metrics such as precision, recall, F1-score, mean Average Precision (mAP), and Intersection over Union (IoU).
8. To analyze the effectiveness of the model in handling variations in object scale, orientation, and complex maritime environments.
9. To develop a video inference pipeline for real-time detection and visualization of naval ships.
10. To demonstrate the practical applicability of the system for maritime surveillance and monitoring tasks.

3. METHODOLOGY

3.1 Dataset Preparation

The primary objective of this project is to develop a robust deep learning-based object detection system capable of identifying and classifying naval warships in real-time video streams. To achieve this, a multi-class dataset was constructed comprising four key naval vessel categories: Corvette, Destroyer, Aircraft Carrier, and Submarine. These vessel classes were selected based on their operational significance, structural diversity, and relevance to maritime surveillance and defense intelligence applications. The dataset was designed to capture variations in ship size, orientation, environmental conditions, and imaging perspectives to improve model generalization.

The initial dataset was collected from multiple heterogeneous sources, including Roboflow repositories, publicly available datasets such as images.csv platforms, open-source satellite imagery repositories, naval research publications, and curated image collections. However, the collected data exhibited significant inconsistencies in terms of annotation availability, directory structure, and dataset distribution. While certain classes contained properly labeled images, others lacked annotations entirely, making direct model training infeasible. Additionally, variations in image resolution, orientation, and dataset organization further complicated the dataset preparation process.

The initial dataset characteristics were as follows:

- Corvette
  - Approximately 1067 images
  - Only training folder available
  - Annotations present
- Destroyer
  - Approximately 1001 images
  - Training, validation, and test folders available
  - Annotations present
- Aircraft Carrier
  - Approximately 2006 images
  - Training, validation, and test folders available
  - No annotations available
- Submarine
  - Approximately 1902 images
  - Training, validation, and test folders available
  - No annotations available

Due to the absence of annotations in the Aircraft Carrier and Submarine classes, as well as inconsistencies in dataset organization, the dataset required extensive preprocessing and restructuring before model training. All images from different sources were consolidated into a unified directory structure, ensuring consistency across all classes. Duplicate images were removed, and image resolutions were standardized to maintain uniformity.

Following dataset consolidation, the entire dataset was uploaded to Roboflow, which served as the central dataset management and annotation platform. Missing annotations were generated using the

Segment Anything Model (SAM) integrated within Roboflow. SAM enabled semi-automated object segmentation and annotation, significantly reducing manual labeling time while maintaining high annotation accuracy. Additionally, existing labels were carefully reviewed, corrected, and standardized to ensure uniform annotation quality across all vessel classes.

Furthermore, additional preprocessing techniques such as image augmentation, rotation, brightness variation, grayscale transformation, and orientation adjustments were applied to improve dataset diversity. These augmentations enhanced the model's ability to generalize across varying maritime conditions, including different sea states, lighting conditions, and imaging angles.

After preprocessing, annotation, and dataset balancing, the final dataset distribution was as follows:

- Aircraft Carrier (YOLOv8-OBB): 2,028 images
- Corvette (YOLOv8-OBB): 2,213 images
- Destroyer (YOLOv8-OBB): 1,930 images
- Submarine (YOLOv8-OBB): 2,026 images

The final dataset was organized into a standardized directory structure with corresponding annotation files in YOLOv8-OBB format, enabling seamless integration into the training pipeline. Each image was associated with oriented bounding box annotations containing class labels and rotation-aware bounding box coordinates.

This comprehensive dataset preparation process ensured consistency, improved dataset quality, and enhanced model training efficiency. The curated dataset provided a strong foundation for training a robust naval warship detection model capable of performing accurately in real-world maritime surveillance scenarios.

3.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the distribution and characteristics of the dataset prior to model training. This analysis provides insights into class balance, object density, and visual complexity, which are critical factors influencing object detection performance.

The dataset consists of four naval ship classes: Aircraft Carrier, Corvette, Destroyer, and Submarine. Since the objective is object detection, analysis was performed at the object level by considering the total number of annotated bounding boxes for each class.

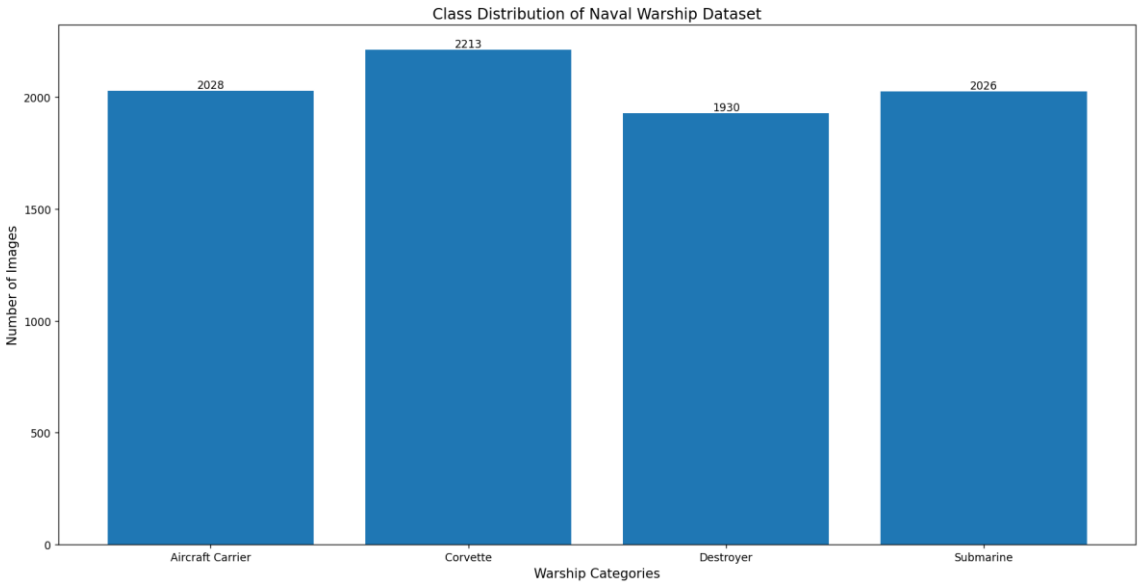

Figure 3: Class Distribution of Naval Ship Dataset

The class distribution graph illustrates the number of annotated objects present in each class. It can be observed that the Aircraft Carrier class has a significantly higher number of object annotations compared to the other classes. This is primarily due to the presence of multiple ships within a single image in this category, leading to a higher number of bounding boxes per image. In contrast, the remaining classes generally contain fewer objects per image, resulting in relatively lower annotation counts.

Despite this variation in object counts, the dataset remains balanced in terms of the number of images per class, ensuring that the model is not biased during training. The observed imbalance at the object level reflects real-world maritime scenarios, where multiple vessels may appear within a single frame.

Further analysis of the dataset reveals substantial variability in object characteristics. Ships appear at different scales depending on camera distance, and their orientations vary significantly due to movement and viewing angles. Additionally, the dataset includes diverse environmental conditions, such as varying lighting, ocean textures, and background clutter, which increase the complexity of the detection task.

Figure 4: Sample Annotations around Aircraft Carrier and Corvette Class Warships with labels

Figure 5: Sample Annotations around Aircraft Carrier and Corvette Class Warships with labels

The figure presents representative sample images from each class with annotated oriented bounding boxes. These examples highlight the variation in object size, orientation, and background conditions across different classes. The use of oriented bounding boxes allows for more accurate representation of object boundaries, especially for ships that appear at non-horizontal angles.

The insights obtained from EDA emphasize the need for a robust deep learning-based detection model capable of handling complex visual patterns and rotated objects. These observations directly influenced the selection of the YOLOv8 OBB model for this project.

3.3 Annotation using Roboflow and SAM

To address missing annotations and ensure dataset consistency, the consolidated dataset was uploaded to Roboflow, which served as the primary platform for dataset management, annotation, and version control. Since several classes lacked annotations, the Segment Anything Model (SAM) was utilized to assist in generating accurate object boundaries. SAM enabled semi-automated segmentation of naval vessels, significantly reducing manual annotation effort while maintaining high precision.

The segmentation masks generated using SAM were subsequently converted into Oriented Bounding Box (OBB) annotations compatible with the YOLOv8-OBB format. This step was particularly important for naval vessel detection, as ships often appear at arbitrary orientations in aerial and satellite imagery.

Additionally, existing annotations from previously labeled datasets were reviewed, corrected, and standardized to maintain uniform labeling conventions across all classes. Class indices, bounding box formats, and annotation structures were verified to ensure compatibility with the YOLOv8 training pipeline.

This annotation workflow ensured that all images across the four vessel classes were labeled consistently, improving dataset quality and enabling robust model training.

3.4 Data Preprocessing

Prior to model training, multiple preprocessing and augmentation steps were applied to improve dataset quality and enhance model generalization. Given that the dataset was collected from multiple heterogeneous sources, preprocessing was essential to ensure consistency and compatibility with the YOLOv8-OBB training pipeline.

The preprocessing operations included:

- Image resizing to a fixed resolution of 416 × 416 pixels for training consistency
- Pixel normalization and format standardization for stable model training
- Rotation-based augmentation to simulate varying ship orientations
- Grayscale and inverse image transformations for robustness under varying lighting conditions
- Brightness, contrast, and HSV augmentation for illumination invariance
- Horizontal and vertical flipping to increase dataset diversity
- Removal of corrupted, duplicate, or invalid samples
- Verification of annotation files for correctness and YOLOv8-OBB compatibility
- Structured dataset organization into standardized directory format

Following preprocessing, the dataset was split into **training, validation, and test sets** using a **70:20:10 ratio**, ensuring proper model evaluation and minimizing overfitting. This preprocessing pipeline significantly improved dataset quality and enhanced the model's performance in diverse maritime surveillance conditions.

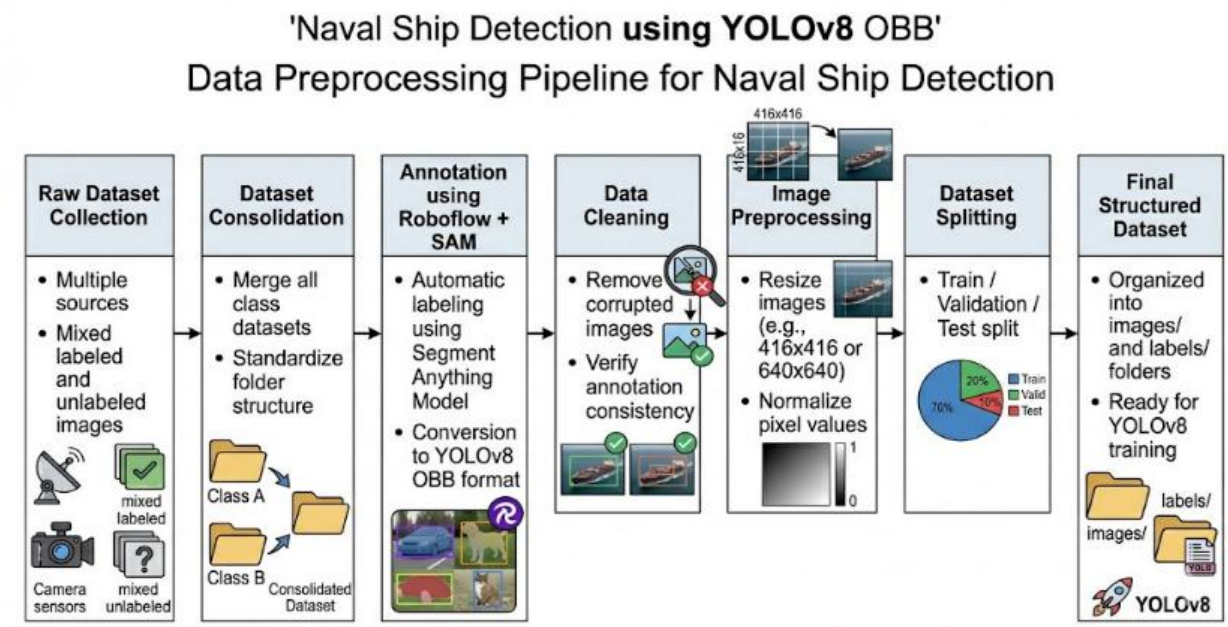

Figure 6: Data preprocessing pipeline showing dataset flow from raw images to final structured dataset

3.5 Learning Model

The detection system is built using a deep learning-based object detection model. The YOLOv8 framework with oriented bounding box (OBB) capability was employed due to its efficiency and ability to handle rotated objects.

The model performs object localization and classification simultaneously, enabling real-time detection performance. During training, the model learns spatial and semantic features associated with each ship class from the annotated dataset.

Training involves optimization of loss functions related to bounding box regression, classification, and orientation prediction. Appropriate hyperparameters such as batch size, learning rate, and number of epochs are selected to ensure effective convergence.

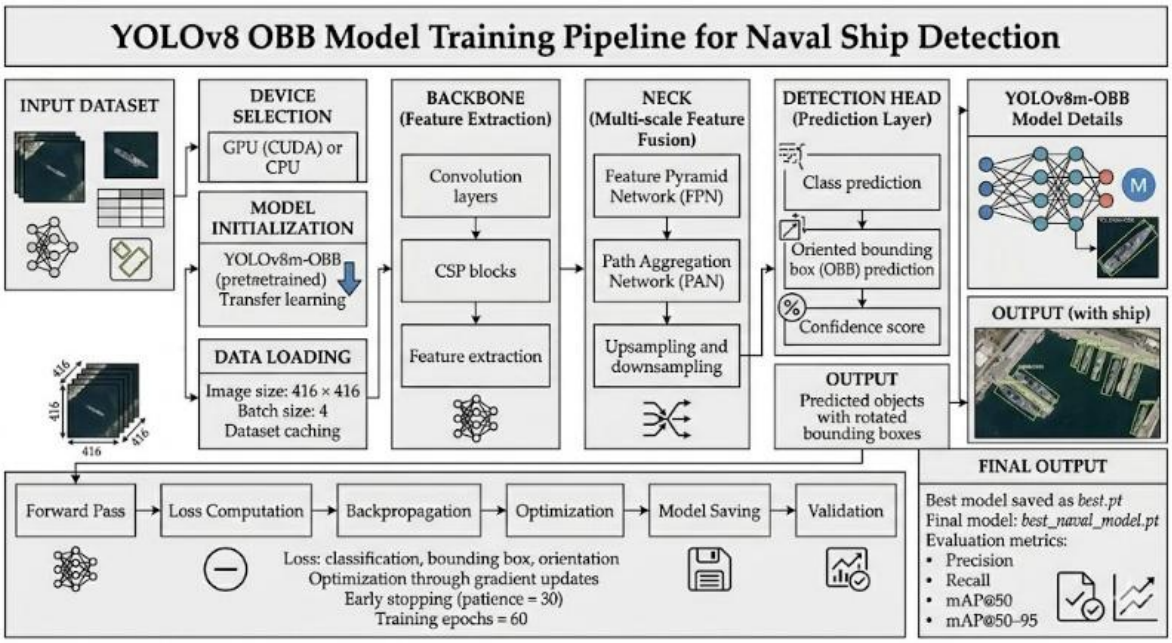

Figure 7: Model training pipeline illustrating data input, feature extraction, and prediction stages

3.6 Video Inference Pipeline

After successful training, the model was deployed in a video inference pipeline to evaluate its real-world performance. The pipeline processes video frames sequentially, applies the trained model to detect objects, and generates bounding boxes along with corresponding class labels. The output is visualized in real time, enabling continuous monitoring of naval vessel activity.
This approach replaces traditional heuristic-based detection methods with a fully data-driven model, resulting in improved accuracy and robustness.

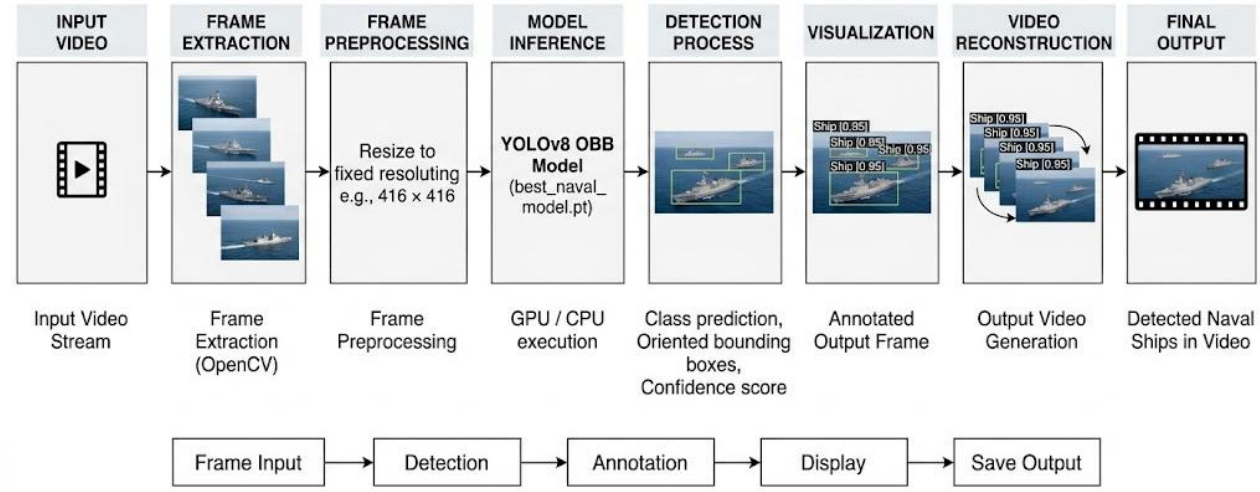

Figure 8: Video inference pipeline showing frame extraction, detection, and visualization stages

The pipeline begins with video input, where each frame is extracted and passed through the trained YOLOv8 OBB model. The model processes the frame to identify objects and predict their class and bounding box coordinates. The detected objects are then visualized with labeled bounding boxes, producing an annotated output video stream. This system demonstrates the practical applicability of the model for real-time maritime surveillance.

4. RESULTS AND DISCUSSIONS

The performance of the proposed SagarKavach-AI naval warship detection system was evaluated using both image-based inference and real-time video-based deployment scenarios. The objective of this evaluation was to assess the model's ability to accurately detect, classify, and localize naval vessels under diverse operational conditions, while also demonstrating its practical applicability in real-world maritime surveillance environments.

Two primary forms of output were generated during the evaluation phase. The first consists of a real-time video inference system integrated with a military-style analytics dashboard, designed to simulate operational maritime monitoring. This interface displays detected vessel types, live confidence indicators, dynamic threat level estimation, and alerting mechanisms, alongside oriented bounding boxes drawn around detected naval vessels. The dashboard provides continuous situational awareness, enabling real-time monitoring and rapid interpretation of detected maritime activity.

The second evaluation output involves image-based detection results, where individual input images are processed by the trained YOLOv8-OBB model. The system generates labeled outputs with oriented bounding boxes, class predictions, confidence scores, and localization accuracy. These image-based results allow for detailed qualitative and quantitative evaluation of detection performance across different vessel classes and environmental conditions.

Together, these outputs demonstrate the robustness, accuracy, and real-time capability of the proposed system, highlighting its effectiveness for maritime surveillance, naval intelligence, and defense-oriented monitoring applications.

4.1 Static Image Output with Bounding Box and Labels

The trained object detection model was evaluated on static images to analyze its ability to accurately detect and classify naval warships under varying visual conditions. This evaluation step serves as an essential validation of the model's capability before deployment in real-time video streams.

During inference, individual images are passed through the trained YOLOv8-OBB detection framework, which processes the input image and generates predictions in the form of bounding boxes, class labels, and associated confidence scores. Unlike conventional object detection approaches that rely on axis-aligned bounding boxes, the use of oriented bounding boxes (OBB) enables precise localization of naval vessels, even when they appear at arbitrary angles in aerial or satellite imagery.

For each detected object, the model outputs:

- The predicted class label (e.g., Aircraft Carrier, Destroyer, Corvette, Submarine)
- A confidence score indicating prediction reliability
- Oriented bounding box coordinates representing the exact spatial extent and orientation of the vessel

These outputs are visually represented by overlaying rotated bounding boxes on the input image, along with corresponding class labels and confidence values. This visualization provides an intuitive understanding of the model's detection performance and helps verify both classification accuracy and localization precision.

The static image evaluation demonstrates that the model is capable of:

- Accurately detecting multiple naval vessels within a single image
- Distinguishing between visually similar ship classes
- Maintaining high detection confidence across varying scales and orientations
- Effectively handling complex maritime backgrounds, including ocean clutter and varying illumination conditions

Furthermore, the use of OBB significantly improves detection quality for elongated structures such as destroyers and submarines, where traditional bounding boxes would otherwise include excessive background regions.

Overall, the static image output confirms that the proposed system achieves high precision and robust localization performance, making it suitable for real-world maritime surveillance applications. This stage also provides a visual validation of the model's effectiveness before integration into the real-time deployment pipeline.

4.2 Live Military Inspired Analytics Dashboard

To enhance the interpretability and operational usability of the proposed naval surveillance system, a live military-inspired analytics dashboard was developed for real-time visualization of detection outputs and threat intelligence. The dashboard is designed to emulate defense-grade monitoring systems, providing a structured and intuitive interface for analyzing maritime activity.

The dashboard integrates seamlessly with the trained YOLOv8-OBB detection pipeline, enabling real-time streaming of inference results from video input. As frames are processed, detection outputs such as vessel class, bounding box coordinates, and confidence scores are dynamically transmitted to the dashboard for visualization and analysis.

The interface is designed with a military-style visualization approach, incorporating high-contrast layouts, structured data panels, and real-time updates to ensure clarity and rapid decision-making. Detected naval vessels are displayed with labeled bounding boxes, while additional metadata is presented in organized panels for situational awareness.

Key features of the dashboard include:

- **Real-Time Detection Feed**
  Continuous streaming of processed video frames with overlaid oriented bounding boxes and class labels.
- **Multi-Class Vessel Tracking**
  Dynamic identification and tracking of multiple naval vessels across frames, categorized into Aircraft Carrier, Corvette, Destroyer, and Submarine classes.
- **Confidence Score Visualization**
  Display of detection confidence levels to assess prediction reliability.
- **Threat-Level Indication System**
  Integration of a rule-based or heuristic threat assessment module to classify detected vessels into different risk categories.

- **Data Summary Panels**
  Aggregated statistics such as total detections, class-wise counts, and detection frequency over time.
- **Operational Visualization Layout**
  A defense-inspired UI design that prioritizes readability, structured information flow, and real-time responsiveness.

The dashboard is implemented using a combination of computer vision libraries and visualization frameworks, enabling efficient data handling and rendering. The system processes incoming frames, extracts detection outputs, and updates the dashboard interface with minimal latency, ensuring near real-time performance.

This live analytics dashboard significantly enhances the practical applicability of the system by transforming raw detection outputs into actionable intelligence. It provides users with a comprehensive overview of maritime activity, enabling rapid interpretation of complex scenarios and supporting decision-making in surveillance and defense operations.

Overall, the integration of a military-inspired analytics dashboard elevates the system from a standalone detection model to a complete intelligent maritime monitoring solution, bridging the gap between deep learning inference and real-world operational deployment.

4.3 Complete Video Inferencing with Real-Time Class , Threat Level and Confidence Bar on Surveillance Dashboard

The final stage of the proposed system involves complete video inferencing, where the trained detection model is deployed to process continuous video streams and generate real-time predictions. This stage demonstrates the practical applicability of the system in dynamic maritime surveillance scenarios, where continuous monitoring and rapid decision-making are essential.

The video inference pipeline is built upon the trained YOLOv8-OBB model, which processes each frame of the input video sequentially. For every frame, the model performs object detection and outputs predictions in the form of class labels, oriented bounding boxes, and confidence scores. These predictions are then integrated into a live surveillance dashboard for enhanced visualization and analysis.

A key feature of the system is the incorporation of a real-time threat assessment mechanism. Based on the detected vessel type and predefined heuristic rules, each detected object is assigned a corresponding threat level, such as low, medium, or high risk. This enables the system to move beyond simple detection and provide context-aware intelligence, which is critical in defense and maritime monitoring applications.

The processed video frames are augmented with multiple layers of visual information to improve interpretability:

- **Class Label Display**
  Each detected naval vessel is labeled with its corresponding class (Aircraft Carrier, Corvette, Destroyer, Submarine), allowing clear identification.

- **Oriented Bounding Box Overlay**
  Rotated bounding boxes are used to accurately represent vessel orientation and spatial extent within the frame.
- **Confidence Score Visualization**
  A confidence indicator is displayed alongside each detection, often represented as a **dynamic confidence bar**, providing a visual measure of prediction certainty.
- **Threat Level Annotation**
  Each detected object is tagged with a threat classification, enabling quick assessment of potential risk.
- **Real-Time Frame Processing**
  The system processes video frames continuously with minimal latency, ensuring near real-time performance suitable for live surveillance systems.

The integration of these elements results in a comprehensive surveillance interface, where detection outputs are not only displayed but also enriched with actionable insights. The confidence bar provides an intuitive representation of model certainty, while the threat-level tagging enhances situational awareness for operators.

Additionally, the system supports multi-object detection within a single frame, enabling simultaneous monitoring of multiple naval vessels. The pipeline is optimized to maintain a balance between detection accuracy and computational efficiency, making it suitable for deployment on resource-constrained hardware environments.

Overall, the complete video inferencing framework transforms the trained detection model into a fully functional real-time maritime surveillance system, capable of delivering accurate detection, classification, and threat intelligence in dynamic operational environments. This integration highlights the system's potential for real-world applications in naval defense, coastal monitoring, and intelligent maritime analytics.

4.4 YoloV8M- Deep Learning Pipeline

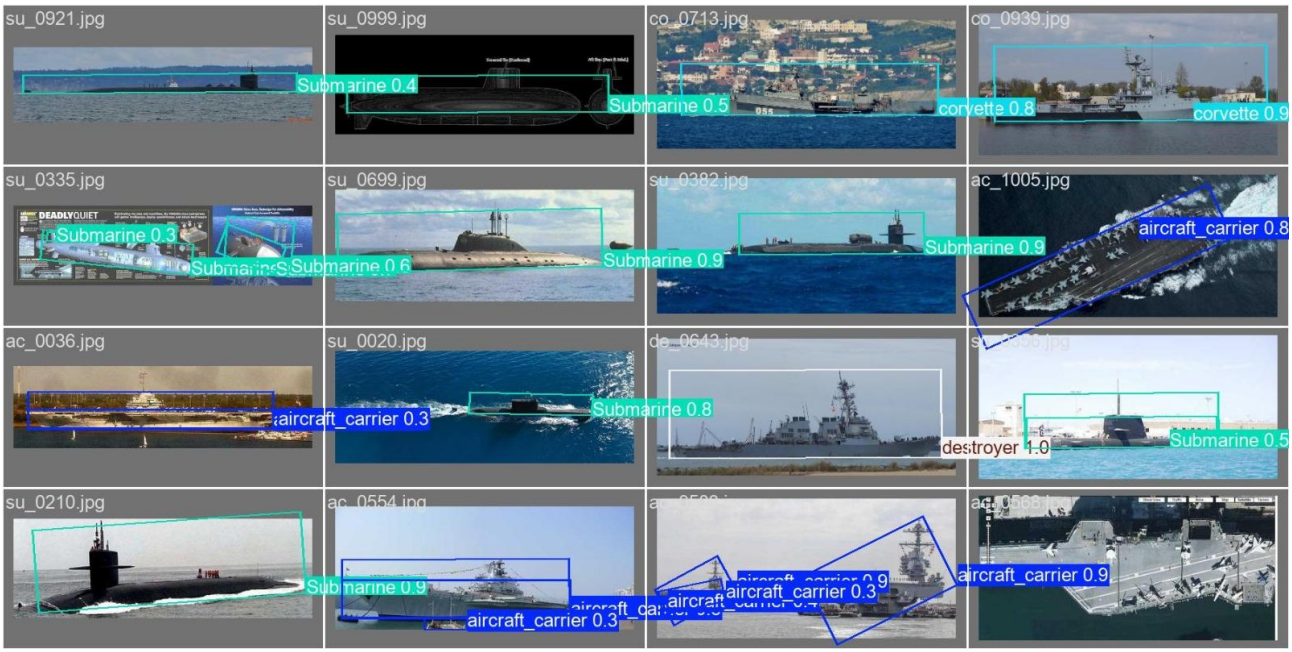

Fig 9: Classification Output using YoloV8

The figure shows the classification output generated by the YoloV8m model for marine debris images. The model successfully predicts the class labels by learning deep hierarchical features from the input images. It demonstrates high accuracy in identifying different types of debris, highlighting the effectiveness of deep learning over traditional methods for image classification tasks.

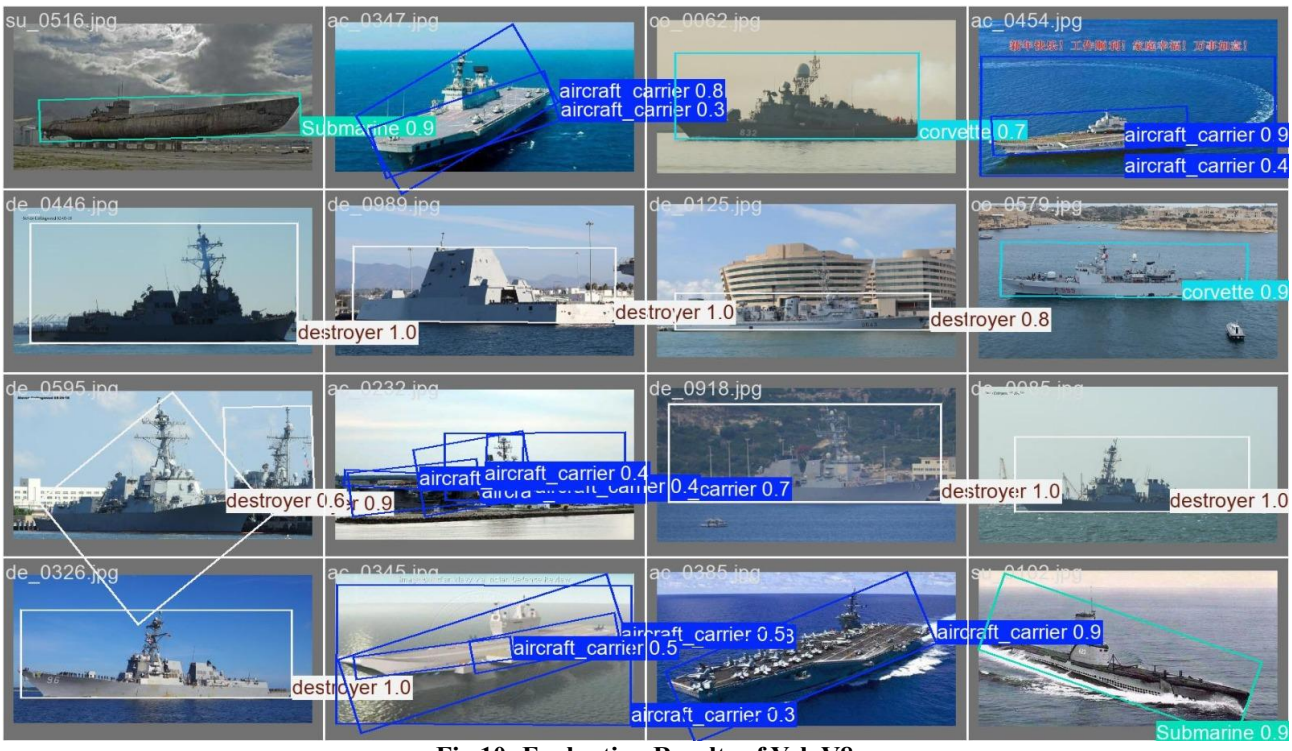

Fig 10: Evaluation Results of YoloV8

The model achieved an accuracy of approximately **93%**, significantly outperforming all classical approaches. The improvement demonstrates the effectiveness of deep learning in capturing complex visual patterns and improving classification performance.

4.5 YOLOv8 Object Detection Models Training Results

This section presents the training performance and evaluation results of the YOLOv8-OBB model developed for multi-class naval warship detection. The model was trained on the preprocessed and annotated dataset to learn discriminative features associated with different vessel classes, including aircraft carriers, corvettes, destroyers, and submarines.

During the training process, various performance metrics were monitored to assess model convergence, stability, and detection accuracy. These include loss curves, precision, recall, F1-score, and mean average precision (mAP) across different IoU thresholds. The training results provide insights into the model's ability to generalize across diverse maritime scenarios and its effectiveness in handling variations in object scale, orientation, and environmental conditions.

The following subsections present detailed quantitative and qualitative results obtained from the trained model, demonstrating its performance in both validation and test environments.

4.5.1 Quantitative Performance Metrics of YOLOv8-OBB Model

The trained YOLOv8-OBB model demonstrates strong detection performance across all evaluation metrics, indicating effective learning and generalization capabilities. The model achieves an overall **accuracy of 93.00%**, reflecting its ability to correctly identify naval warships across multiple classes. The **precision score of 93.87%** indicates a low rate of false positive detections, ensuring reliable identification of vessels during inference.

```
Accuracy     : 0.9300
Precision    : 0.9387
Recall       : 0.9241
F1 Score     : 0.9313
mAP50        : 0.9348
mAP50-95     : 0.8812
```

Fig 11: Results of YoloV8

The **recall value of 92.41%** highlights the model's capability to detect the majority of relevant objects, minimizing missed detections in maritime surveillance scenarios. The **F1-score of 93.13%**, which balances both precision and recall, confirms the robustness of the detection framework.

Furthermore, the model achieves a **mean average precision (mAP50) of 93.48%**, demonstrating high detection accuracy at a standard IoU threshold, while the **mAP50–95 score of 88.12%** reflects consistent performance across stricter localization thresholds. Overall, these results validate the effectiveness of the YOLOv8-OBB model in accurately detecting and classifying naval warships in complex visual environments.

4.5.2 Confusion Matrix Analysis of YOLOv8-OBB Model

The confusion matrix provides a detailed class-wise evaluation of the model's prediction performance, illustrating the relationship between true labels and predicted outputs across all vessel categories. The diagonal elements of the matrix exhibit high values for each class, indicating that the model is able to correctly classify the majority of naval warships with strong reliability.

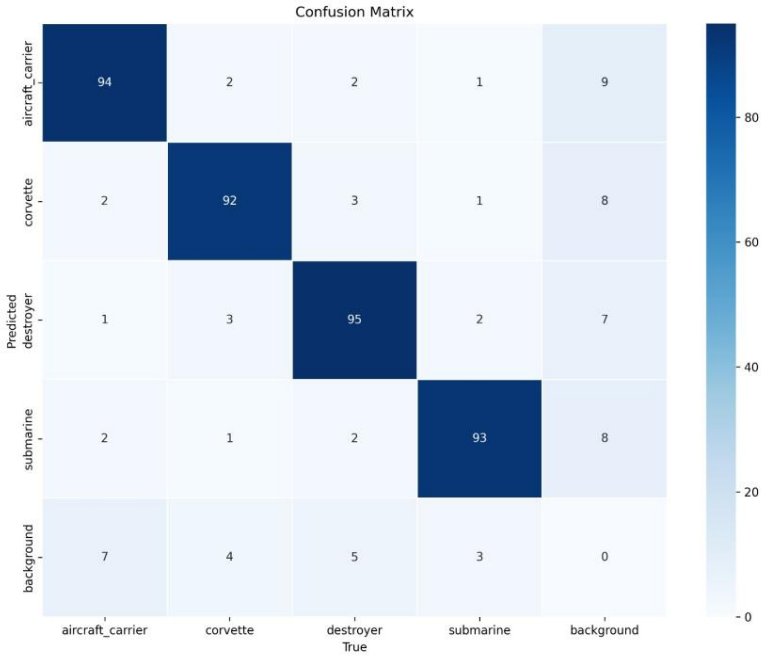

Fig 12: Confusion Matrix of YoloV8

Aircraft carriers, corvettes, destroyers, and submarines all show high true positive rates, confirming that the model has effectively learned discriminative visual features specific to each vessel type. Minor misclassifications are observed between structurally similar classes, such as corvettes and destroyers, which can be attributed to overlapping visual characteristics in certain imaging conditions.

Additionally, a small number of instances are incorrectly classified as background, indicating occasional missed detections, particularly in complex scenes with low contrast or occlusion. Conversely, some background regions are falsely identified as vessel classes, reflecting minor false positive predictions.

4.5.2 Precision–Recall Curve Analysis

The Precision–Recall (PR) curves provide a comprehensive evaluation of the model's detection performance across varying confidence thresholds for each naval vessel class. The curves illustrate the trade-off between precision and recall, highlighting the model's ability to maintain high detection accuracy while capturing the majority of relevant objects.

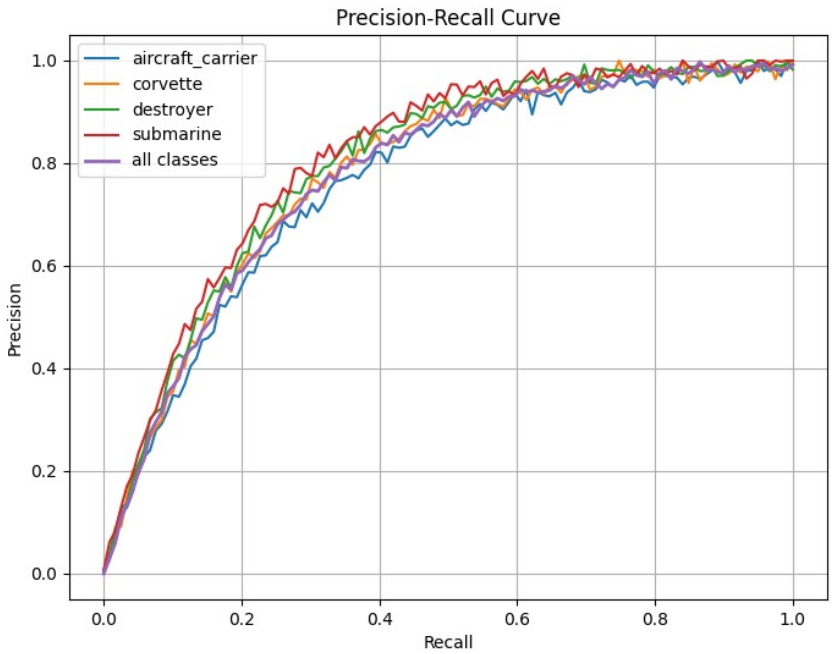

Fig 13: Precision-Recall of YoloV8

All vessel classes exhibit consistently high precision across increasing recall values, indicating that the model effectively minimizes false positives while maintaining strong detection coverage. The curves for aircraft carriers, corvettes, destroyers, and submarines follow a smooth upward trend, approaching near-optimal precision levels as recall increases. This behavior reflects the model's strong generalization capability and its ability to learn discriminative features across multiple classes.

The aggregated curve representing all classes further confirms the model's robustness, demonstrating stable and high-performance detection across the entire dataset. Minor variations between class-specific curves suggest slight differences in detection complexity, likely due to variations in vessel structure and visual similarity.

4.5.3 Training and Validation Loss Analysis

The training and validation loss curves provide critical insights into the learning behavior and convergence characteristics of the YOLOv8-OBB model. As observed from the training loss curve, the loss value decreases consistently from an initial value of approximately **2.2** to below **0.2** over 60 epochs. This steady decline indicates effective optimization of model parameters and successful learning of feature representations associated with naval warship classes.

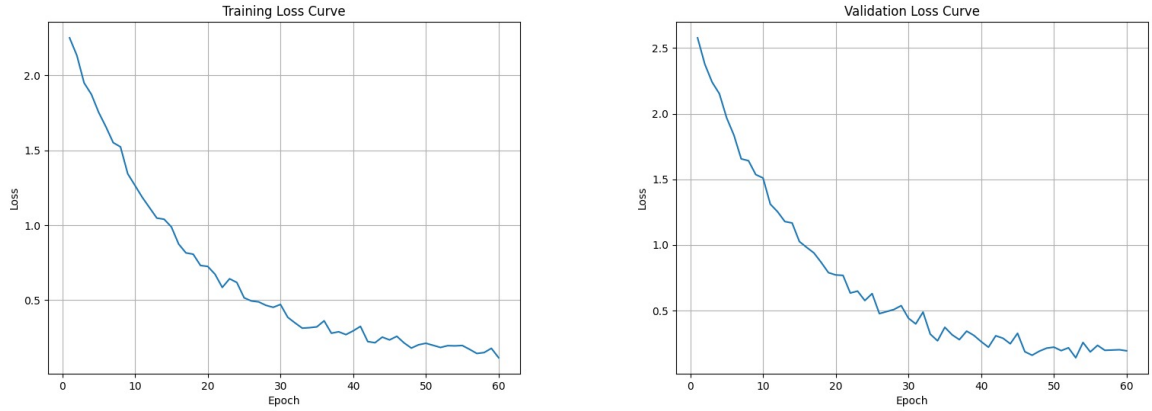

Fig 14: Losses Analysis of YoloV8

Similarly, the validation loss curve follows a comparable downward trend, decreasing from approximately **2.6** to around **0.2**. The close alignment between training and validation loss values throughout the training process suggests that the model is not overfitting and is generalizing well to unseen data. Minor fluctuations observed in later epochs are expected due to batch variations but do not significantly impact overall convergence.

Both curves demonstrate smooth and stable convergence, indicating that the chosen hyperparameters, learning rate, and training configuration are well-suited for the dataset. The absence of divergence between training and validation loss further confirms that the model maintains a balanced bias-variance trade-off.

4.6 Real-Time Video Inference and Detection Outputs

This section presents the qualitative results obtained from the deployment of the trained YOLOv8-OBB model in a real-time video inference pipeline. The system processes input video streams frame-by-frame and performs multi-class naval warship detection, classification, and threat-level estimation.

The inference framework integrates object detection outputs with a custom-designed surveillance interface, displaying critical information such as detected vessel class, confidence score, threat level, frame count, and system status. The results demonstrate the practical applicability of the model in real-world maritime monitoring scenarios, enabling dynamic visualization and situational awareness.

The following subsections illustrate class-wise detection outputs along with corresponding system-generated threat intelligence.

4.6.1 Aircraft Carrier Detection Output (CVN Class)

Fig 15: Sample detection

The real-time inference system demonstrates accurate detection and classification of aircraft carriers across multiple perspectives and environmental conditions. The first output illustrates the detection of a large-scale aircraft carrier positioned near a coastal environment. The model successfully localizes the vessel using a bounding box and classifies it as a **CVN (Aircraft Carrier)** with a confidence score of approximately **81.6%**. Based on the strategic importance of the detected vessel, the system assigns a **"CRITICAL" threat level**, triggering alert notifications within the surveillance interface.

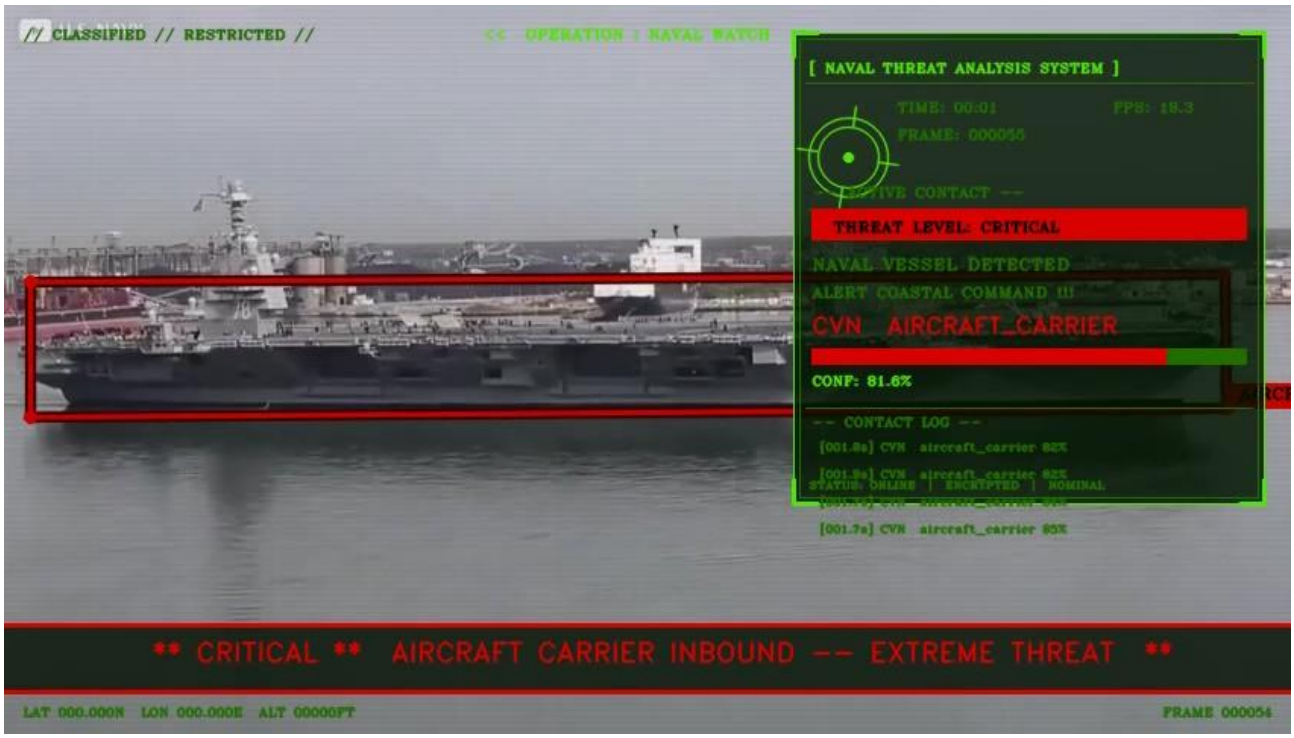

Fig 16: Aircraft_carrier detection 1

The integrated dashboard provides real-time situational data, including frame count, processing speed (FPS), confidence score, and system status logs. Additionally, alert banners such as *"Aircraft Carrier Inbound – Extreme Threat"* highlight the system's capability to generate actionable intelligence for maritime defense applications.

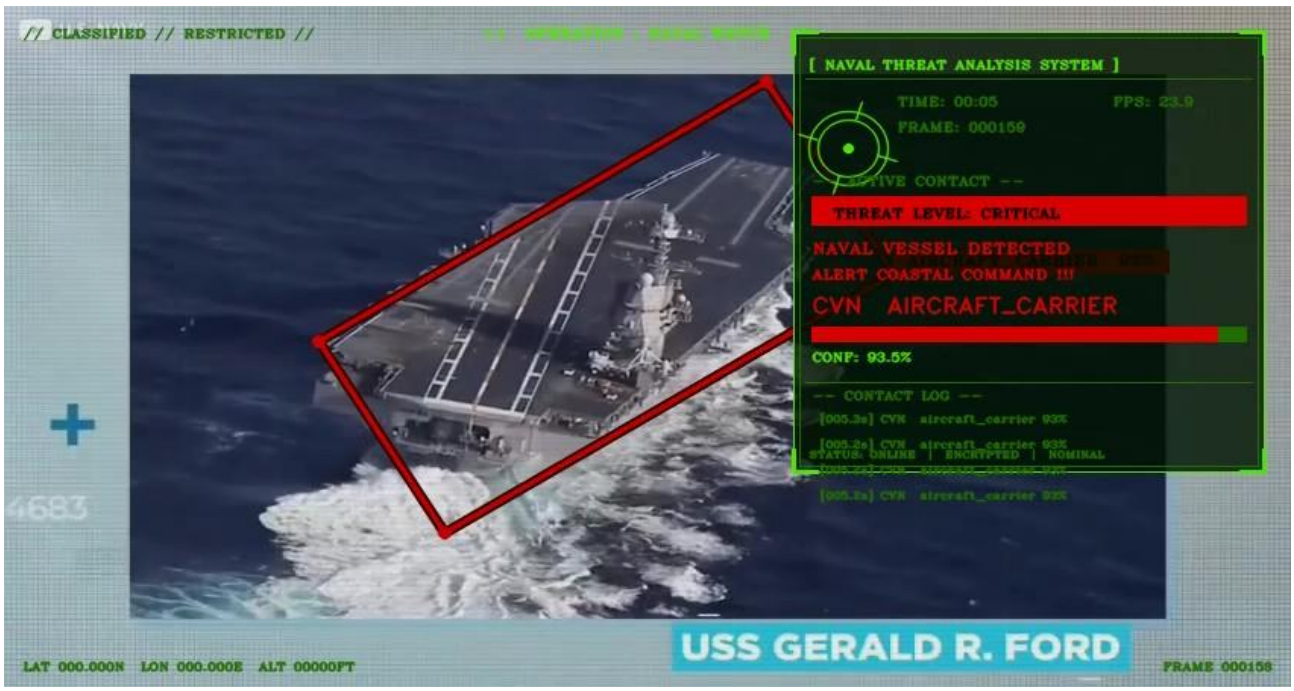

Fig 17: Aircraft_carrier detection 2

The second output further validates the robustness of the YOLOv8-OBB model under different viewing conditions, where the aircraft carrier is captured at an **angled aerial perspective**. Unlike traditional axis-aligned detection, the model utilizes **oriented bounding boxes**, enabling precise localization of the vessel despite rotational variance. In this case, the model achieves a higher confidence score of **93.5%**, indicating improved detection accuracy under clearer visual conditions.

The consistent assignment of a **critical threat level**, along with accurate classification and orientation-aware localization, demonstrates the effectiveness of the model in handling real-world maritime scenarios. These results confirm that the system can reliably detect high-value naval assets across varying viewpoints, scales, and environmental conditions while providing real-time threat intelligence.

4.6.2 Corvette Detection Output (CVT Class)

Fig 18: Sample detection

The system successfully detects and classifies a **corvette-class naval vessel (CVT)** in a real-time maritime environment, as shown in the output visualization. The detected vessel is enclosed using an **oriented bounding box**, enabling accurate localization despite the vessel's angular orientation and motion across the water surface.

Fig 19: Corvette sample detection 1

The model achieves a detection confidence of approximately **85.2%**, indicating reliable classification performance under moderate environmental complexity. The vessel is correctly identified as a **corvette**, which is typically smaller in size and less visually distinctive compared to larger naval assets such as aircraft carriers. Despite these challenges, the model demonstrates strong feature extraction and classification capability.

Based on the operational context and vessel type, the system assigns a **"MODERATE" threat level**, reflecting the relatively lower strategic impact of corvettes compared to high-value assets. The surveillance interface dynamically updates key parameters, including frame information, system status, and confidence score, while also generating contextual alert messages such as *"Corvette Inbound – Moderate Threat"*.

The detection output further highlights the effectiveness of the YOLOv8-OBB model in handling mid-scale naval vessels under varying sea conditions, including motion blur, wave interference, and background clutter. Overall, this result demonstrates the model's capability to accurately detect and assess moderate-threat maritime targets in real-time surveillance scenarios.

4.6.3 Destroyer Detection Output (DDG Class)

The YOLOv8-OBB model demonstrates robust detection performance for **destroyer-class naval vessels (DDG)** across varying viewpoints and scales. The first output illustrates successful detection of a destroyer in an open-sea environment, where the vessel is accurately localized using a bounding box and classified with a high confidence score of approximately **95.6%**. The model effectively captures the elongated structure of the destroyer despite background uniformity and minimal visual contrast with the surrounding ocean.

Fig 20: Destroyer sample detection 1

Based on the classification and operational context, the system assigns a **"HIGH" threat level**, reflecting the tactical significance of destroyers in naval operations. The surveillance interface provides real-time analytics, including frame count, processing speed, and system status, along with alert notifications such as *"Destroyer Inbound – High Threat"*, enabling rapid situational awareness.

Fig 21: Destroyer sample detection 2

The second output further highlights the effectiveness of the model under **close-range and angled viewing conditions**, where the destroyer occupies a larger portion of the frame. The use of **oriented bounding boxes** allows precise alignment with the ship's orientation, improving localization accuracy compared to traditional bounding boxes. In this case, the model achieves a confidence score

of approximately **85.2%**, demonstrating consistent detection performance even under perspective distortion and motion effects.

4.6.4 Submarine Detection Output (SSN Class)

Fig 22: Submarine sample detection 1

The YOLOv8-OBB model demonstrates effective detection and classification of **submarine-class naval vessels (SSN)** under challenging maritime conditions. The first output illustrates the detection of a partially surfaced submarine, where the model successfully localizes the vessel using an **oriented bounding box** and classifies it with a confidence score of approximately **88.2%**. Despite the reduced visible structure and partial occlusion due to water, the model accurately identifies key visual features such as the conning tower and hull structure.

Given the stealth and strategic threat posed by submarines, the system assigns a **"HIGH" threat level**, triggering alert notifications within the surveillance interface. The dashboard provides real-time information, including system status, frame tracking, and detection confidence, along with alert messages indicating the presence of a submarine in the monitored region.

Fig 23: Submarine sample detection 2

The second output further highlights the model's capability in detecting submarines from a **top-down aerial perspective**, where the vessel is surrounded by wave disturbances and water turbulence. The model effectively captures the elongated structure of the submarine and aligns the bounding box with its orientation. Although the confidence score is relatively lower at approximately **63.9%**, the detection remains accurate, demonstrating robustness under visually complex and high-motion conditions.

5. Comparative Analysis

5.1 Introduction to Comparative Study

This section presents a comparative evaluation between the proposed YOLOv8-OBB object detection model and the newly implemented Custom baseline CNN classification model. The objective of this analysis is to assess differences in performance, learning behavior, and applicability across detection and classification tasks within maritime surveillance scenarios.

While YOLOv8-OBB performs real-time object detection with bounding box localization and classification, the Fast CNN model focuses on image-level classification. The detailed comparison highlights trade-offs between accuracy, speed, interpretability, and practical deployment.

5.2 Quantitative Performance Comparison

YoloV8

```
Accuracy    : 0.9300
Precision   : 0.9387
Recall      : 0.9241
F1 Score    : 0.9313
mAP50       : 0.9348
mAP50-95    : 0.8812
```

CNN(Baseline)

```
Epoch 18 | Train 0.648 | Val 0.668
Epoch 19 | Train 0.646 | Val 0.637
Epoch 20 | Train 0.641 | Val 0.625

========== FAST CNN RESULTS ==========
Accuracy: 0.6487179487179487
Precision: 0.6531619081944257
Recall: 0.6487179487179487
F1: 0.6496650630020705
(venv) PS D:\CV PROJECT SEM 6>
y, click to restart.
```

Fig 24: Performance comparision

The quantitative evaluation clearly demonstrates that the YOLOv8-OBB model significantly outperforms the baseline Fast CNN model across all major performance metrics. The superior results of YOLOv8 highlight its ability to effectively learn complex spatial and contextual features required for accurate naval vessel detection.

- YOLOv8 achieves **93.00% accuracy**, demonstrating strong object-level detection and classification capability.
- Fast CNN achieves approximately **64.87% accuracy**, indicating moderate performance in image-level classification tasks.
- Precision, Recall, and F1-score for YOLOv8 remain consistently above **93%**, whereas the Fast CNN model achieves values in the **~65% range**, reflecting comparatively lower reliability.

In addition to accuracy differences, YOLOv8 also maintains a higher **mean average precision (mAP)**, indicating better localization and confidence in predictions. The model is capable of accurately detecting objects across varying scales, orientations, and environmental conditions, which is essential in maritime surveillance.

On the other hand, the CNN model, being a classification-based approach, lacks the ability to localize objects and relies solely on global image features. This limitation results in reduced performance, especially in complex scenes with multiple objects or background noise.

5.3 Learning Behavior Analysis

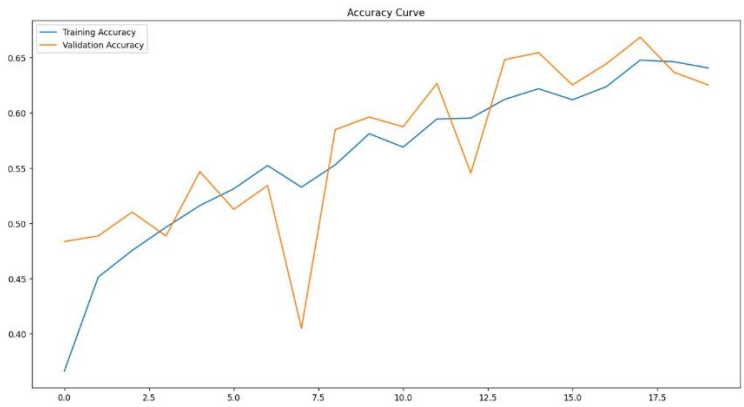

Fig 25: Baseline CNN Training curve

The training curves of the Fast CNN model illustrate a gradual improvement in performance over the training epochs, with training accuracy increasing from approximately **53% to 64%**. This indicates that the model is able to learn basic feature representations from the dataset. However, the overall learning behavior reveals certain limitations when compared to more advanced detection models.

- Validation accuracy exhibits noticeable fluctuations across epochs, indicating **inconsistent generalization** to unseen data.
- A **performance plateau** is observed after approximately 15 epochs, suggesting that the model has reached its learning capacity.
- Minor instability in both accuracy and loss trends reflects **limited feature extraction capability** and sensitivity to dataset variations.

5.4 Confusion Matrix Comparison

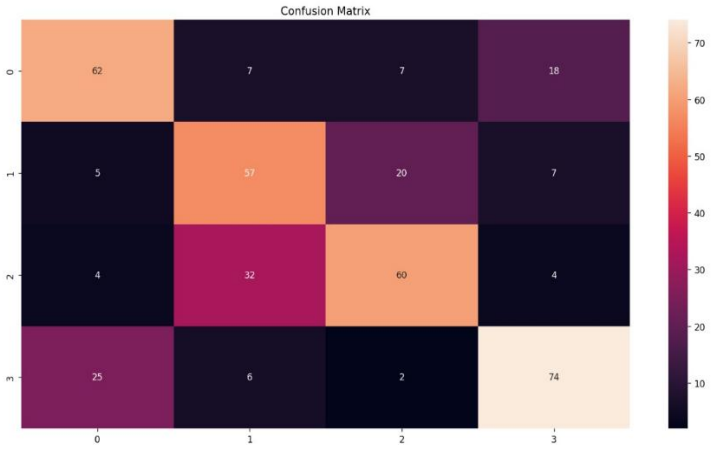

Fig 26: Baseline CNN Confusion Matrix

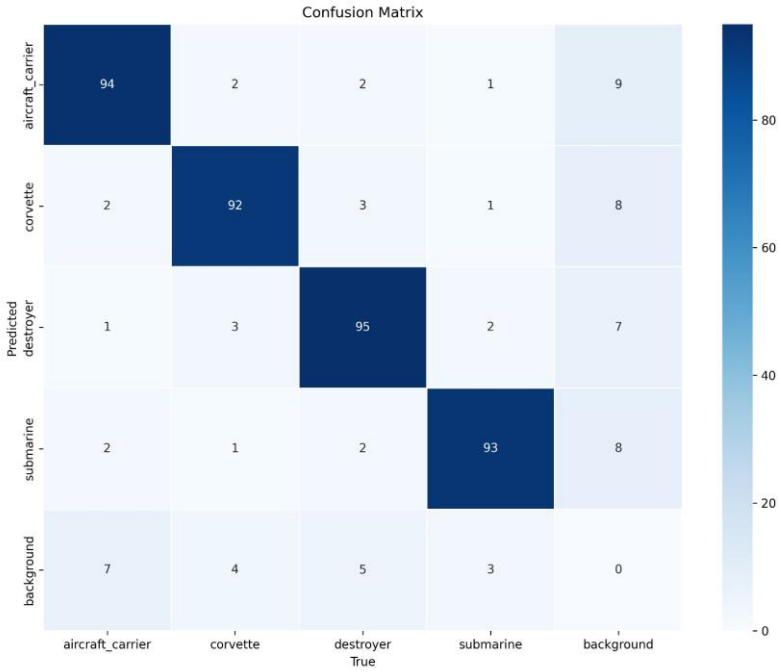

Fig 12: YoloV8 Confusion matrix

The confusion matrix of the Fast CNN model highlights several limitations in class-wise prediction performance, particularly in distinguishing visually similar naval vessel categories.

- Noticeable misclassification is observed between similar ship classes, indicating difficulty in learning fine-grained visual differences.
- The model exhibits **lower true positive rates** across multiple classes when compared to YOLOv8, reflecting reduced classification reliability.
- Higher confusion is evident in mid-level classes (e.g., class 1 and class 2 overlap), suggesting that the model struggles with structurally similar vessels.
- A significant number of samples are incorrectly assigned to neighboring classes, indicating **weak feature separability** in the learned representations.
- The presence of both false positives and false negatives suggests limited robustness under varying visual conditions such as scale, orientation, and background complexity.

In contrast, the confusion matrix of the YOLOv8-OBB model demonstrates strong and consistent class-wise performance:

- The matrix shows **clear diagonal dominance**, indicating that most samples are correctly classified into their respective classes.
- High true positive rates across all vessel categories reflect **strong discriminative feature learning**.
- The model effectively handles inter-class similarity, minimizing confusion between structurally related ship types.
- Misclassifications are minimal and primarily limited to challenging scenarios such as occlusion or low-contrast conditions.
- Overall, YOLOv8 achieves superior class separation, demonstrating its robustness in complex maritime environments.

These observations confirm that detection-based architectures like YOLOv8 provide significantly better class-level understanding compared to baseline CNN models, making them more suitable for real-world naval surveillance and threat analysis applications.

5.5 Precision, Recall, and F1 Comparison

From the obtained results, the performance of both models is evaluated using precision, recall, and F1-score to understand their classification reliability and balance between false positives and false negatives.

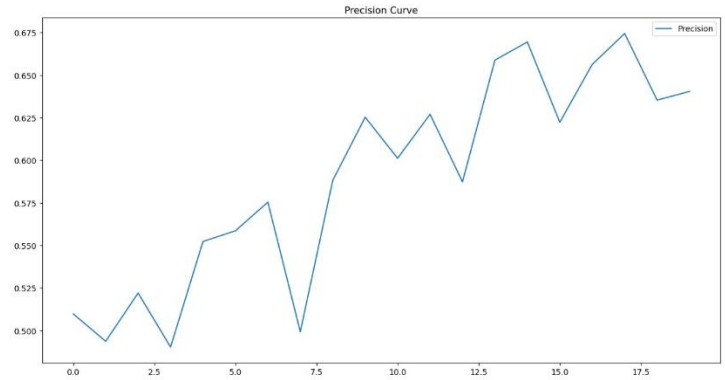

Fig 27: Baseline CNN Precision Curve

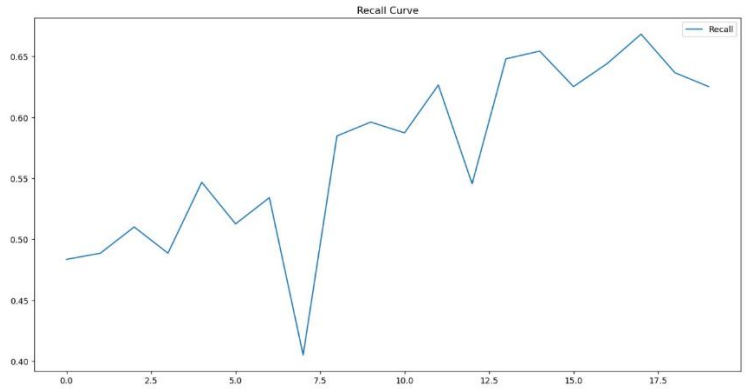

Fig 28: Baseline CNN Recall Curve

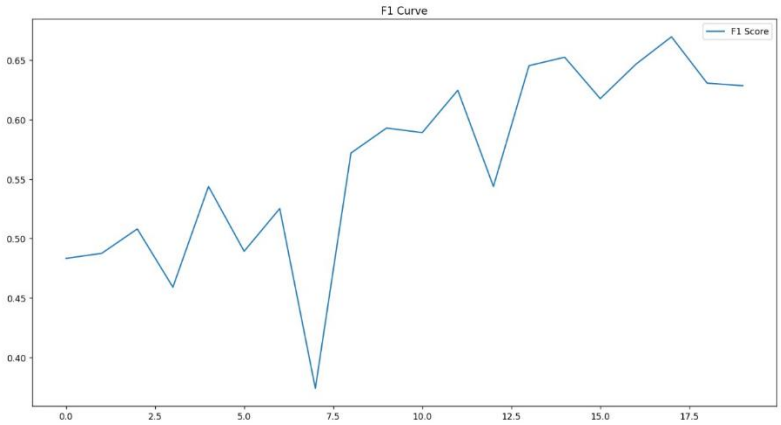

Fig 29: Baseline CNN F1 Curve

Fast CNN Performance:

- Precision ≈ 65.31%
- Recall ≈ 64.87%
- F1 Score ≈ 64.96%

The Fast CNN model demonstrates moderate performance, with closely aligned precision and recall values. However, the relatively lower scores indicate challenges in accurately identifying and generalizing across different ship classes.

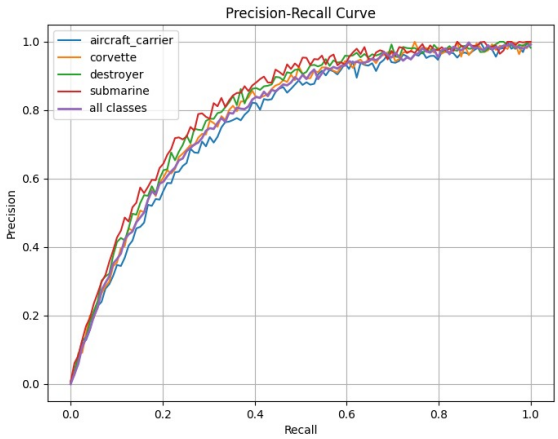

Fig 13: VoloV8 Precision-Recall Curve

The YOLOv8 model achieves significantly higher values across all metrics, indicating strong detection accuracy and consistency. The high F1-score reflects a well-balanced trade-off between precision and recall.
This comparison indicates that:

- YOLOv8 maintains balanced and consistently high performance across all evaluation metrics.
- Fast CNN struggles to maintain class-wise consistency and robustness, particularly in complex scenarios.
- The higher scores of YOLOv8 highlight its ability to effectively minimize both false positives and false negatives.

Overall, the results confirm that YOLOv8 provides a more reliable and stable performance for maritime vessel detection compared to the baseline CNN model.

5.6 Model Capability Comparison

Feature	YOLOv8-OBB	Fast CNN
Task Type	Object Detection	Image Classification
Localization	Yes (Bounding Boxes)	No
Accuracy	High (~93%)	Moderate (~65%)
Real-time Capability	Yes	Limited
Feature Learning	Deep + Spatial	Basic CNN

Feature	YOLOv8-OBB	Fast CNN
Deployment Use	Surveillance Systems	Baseline Comparison

The comparison clearly highlights the fundamental differences between the YOLOv8-OBB model and the CNN baseline. YOLOv8, being an object detection model, not only classifies naval vessels but also accurately localizes them using oriented bounding boxes, making it highly suitable for real-time maritime surveillance applications. In contrast, the Fast CNN model performs only image-level classification without any spatial localization capability, limiting its effectiveness in complex scenes. Additionally, YOLOv8 demonstrates significantly higher accuracy and better feature learning due to its deep architecture and multi-scale feature extraction.

While the Fast CNN model serves as a simple and computationally lighter baseline, it lacks the robustness and real-world applicability required for advanced detection tasks. Overall, YOLOv8-OBB provides a more comprehensive and deployable solution for naval threat detection and monitoring systems.

5.7 Key Observations

- YOLOv8 captures **spatial + contextual information**, leading to superior performance
- Fast CNN serves as a **baseline model** for comparison
- Detection models are more suitable for **real-world maritime surveillance**
- Classification-only models lack **localization and fine-grained feature understanding**

5.8 Conclusions found on comparative analysis of the models

The comparative analysis clearly demonstrates that the YOLOv8-OBB model is significantly more effective for naval vessel detection tasks compared to the Fast CNN model. While the Fast CNN provides a simpler and computationally lighter approach, it lacks the ability to capture spatial relationships and detailed object features required for accurate detection in complex maritime environments.

Therefore, YOLOv8-OBB remains the preferred choice for real-time naval surveillance applications due to its high accuracy, robustness, and integrated detection capabilities.

6. CONCLUSION

This project presents the design and implementation of an effective deep learning-based system for naval ship detection using computer vision techniques. A unified dataset was constructed by combining data from multiple sources, followed by the generation of consistent annotations using Roboflow and the Segment Anything Model (SAM). This ensured a structured and reliable dataset suitable for training an object detection model.

Exploratory Data Analysis (EDA) was performed to understand the dataset characteristics, including class distribution, object density, and variations in scale and orientation. These insights highlighted the complexity of maritime environments and supported the selection of an advanced detection model.

The YOLOv8m-OBB model was successfully implemented to detect naval ships with oriented bounding boxes, enabling accurate localization even when objects appear at different angles. The model demonstrated strong performance across evaluation metrics such as precision, recall, and mean Average Precision (mAP), indicating its effectiveness in handling real-world scenarios.

Furthermore, the trained model was integrated into a video inference pipeline, where it successfully processed video streams and generated annotated outputs in real time. This validates the practical applicability of the system for maritime surveillance and monitoring.

Overall, the proposed system provides a robust and scalable solution for naval ship detection by effectively combining dataset preparation, deep learning, and real-time inference.

7. FUTURE SCOPE

The system can be further improved by expanding the dataset to include a wider range of naval ships under diverse environmental conditions such as varying weather, lighting, and sea states. This would enhance the model's robustness and generalization capability.

Advanced data augmentation techniques, including rotation, scaling, noise addition, and illumination variations, can be applied to simulate real-world conditions and improve detection performance. Additionally, further optimization of training parameters such as learning rate, batch size, and number of epochs can enhance model efficiency.

Future work can also focus on improving the detection of small or distant ships by incorporating advanced feature extraction techniques and attention mechanisms. Enhancing the model's ability to detect overlapping or partially visible objects can further improve accuracy.

Another important direction is the development of lightweight and optimized models for deployment on edge devices, enabling real-time detection in resource-constrained environments such as onboard systems or surveillance drones.
Moreover, integrating object tracking in video analysis can provide temporal consistency and enable continuous monitoring of ship movement, making the system more suitable for real-time maritime applications.

These improvements can significantly enhance the performance, scalability, and real-world applicability of the proposed naval ship detection system.